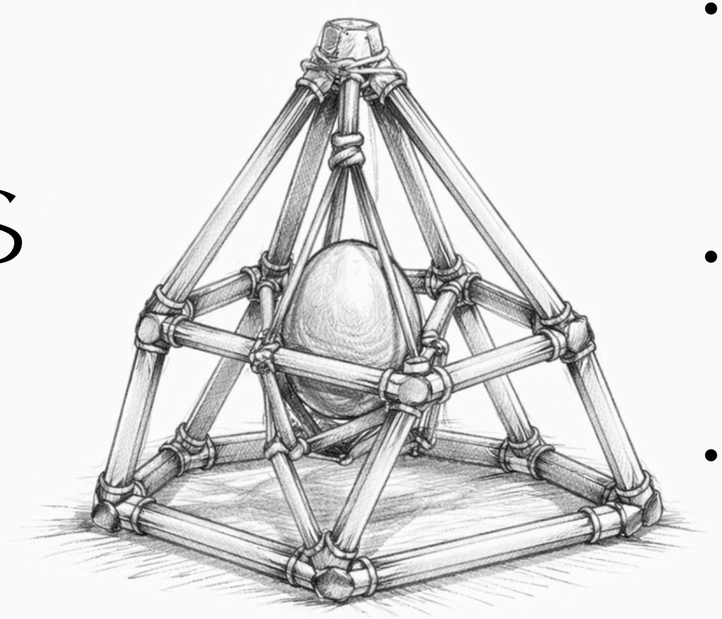
EGG DROP CHALLANGE

IDE 102 NEHİR KARAVAŞ

SURVIVE

THE EGG

KEY DESIGN MILESTONES



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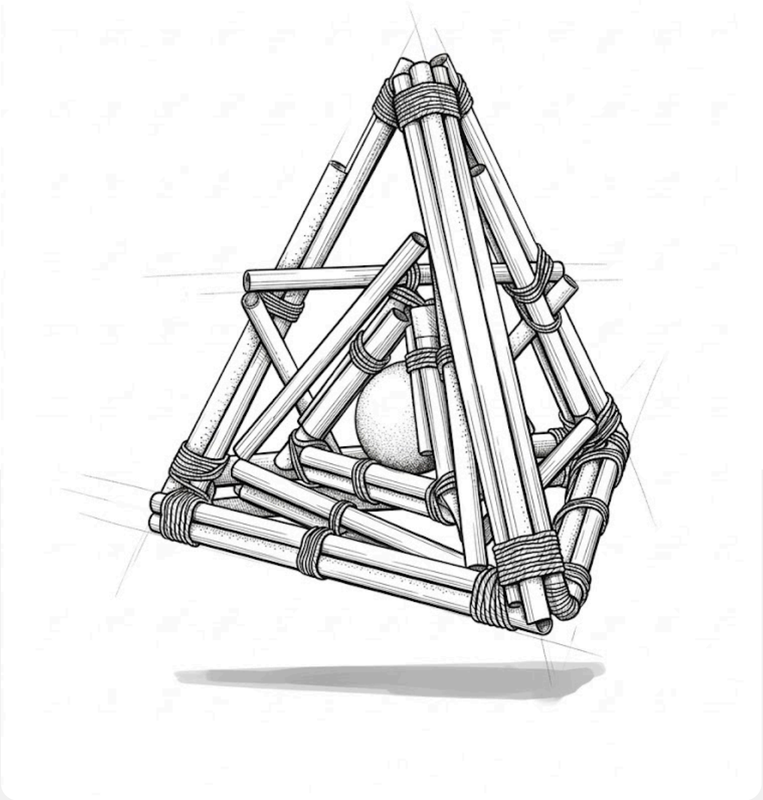
- **Milestone 1:** Construction of a rigid tetrahedral (triangle-based) frame using straws. The goal was structural stiffness, but the results were definitive: the structure was too stiff.
- **Milestone 2 (The Failure):** During a 3-meter drop test, the rigidity of the frame acted as a shock conductor. While the frame survived, the internal shock shattered the egg.
- **Milestone 3 (AI Collaboration):** Troubleshooting G-force failures with AI to shift the design philosophy from "strength" to "energy management."
- **Milestone 4 (Final Execution):** Transitioning to a hexagonal suspension system that successfully balanced the strict material limits with the chaotic forces of gravity.

01.

COMPARE



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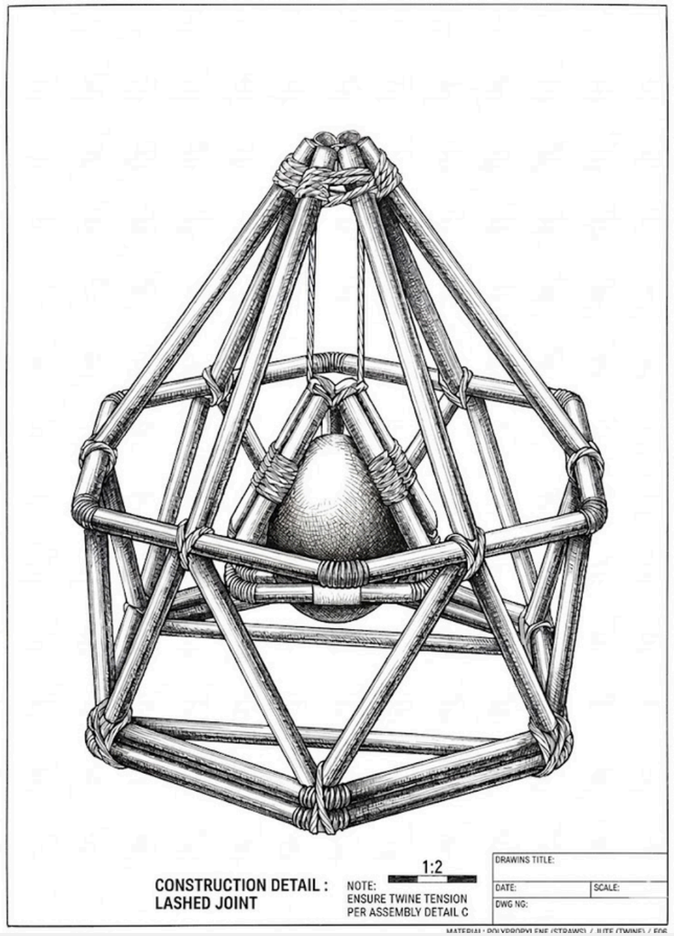
1.5 METER

3 METER



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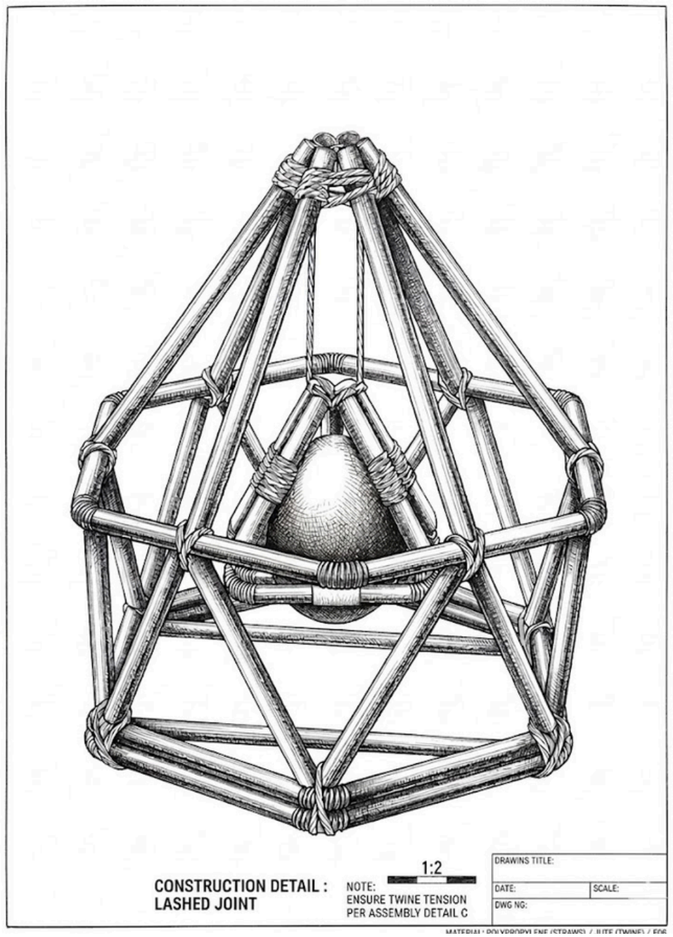
3 METER



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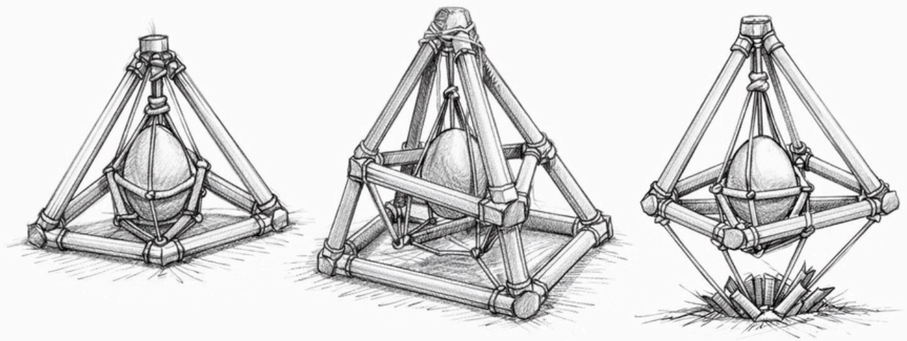
KEYWORDS

- **Kinetic Energy Dissipation:** Converting the energy of a fall into structural flex and movement rather than allowing it to concentrate at the core.
- **Controlled Deformation:** Engineering the base to bend intentionally, acting as a natural spring that consumes impact energy.
- **Decoupled Suspension:** Isolating the payload from the exoskeleton to ensure vibrations are filtered out before reaching the egg.
- **G-Force Mitigation:** Increasing the "deceleration time." By allowing the egg to sway, we lower the peak force experienced by the eggshell.

SELECTED IMPACTFUL PROMPTS

02.

The Prompt: "My first egg drop structure failed because of internal G-force. The egg was suspended inside a rigid straw frame. Can you suggest and sketch simple structural systems that fit inside a 15 cm cube and are buildable with straws and tape? Please focus on kinetic energy dissipation instead of only cushioning." This inquiry was designed to move beyond "building a cage" and toward "engineering a system." It forced a focus on force flow rather than static shapes.



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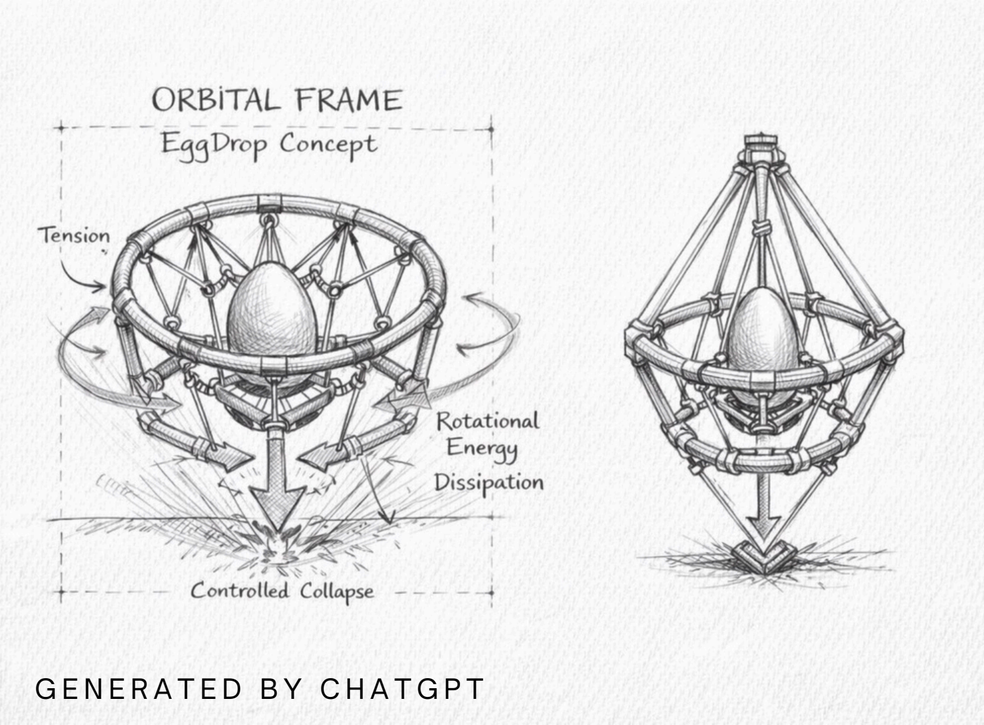
AI ROLE TAGS

- **Clarifier:** Defined the "Rigidity Trap," explaining that my first design failed because it acted as a shock conductor rather than an insulator.
- **Re-framer:** Shifted the challenge from "How do I make it stronger?" to "How do I move the energy away?"
- **Expander:** Introduced 6-point suspension theories and non-linear geometries (hexagons).
- **Critique:** Provided a reality check on joints, warning that excessive tape would increase weight and brittleness.



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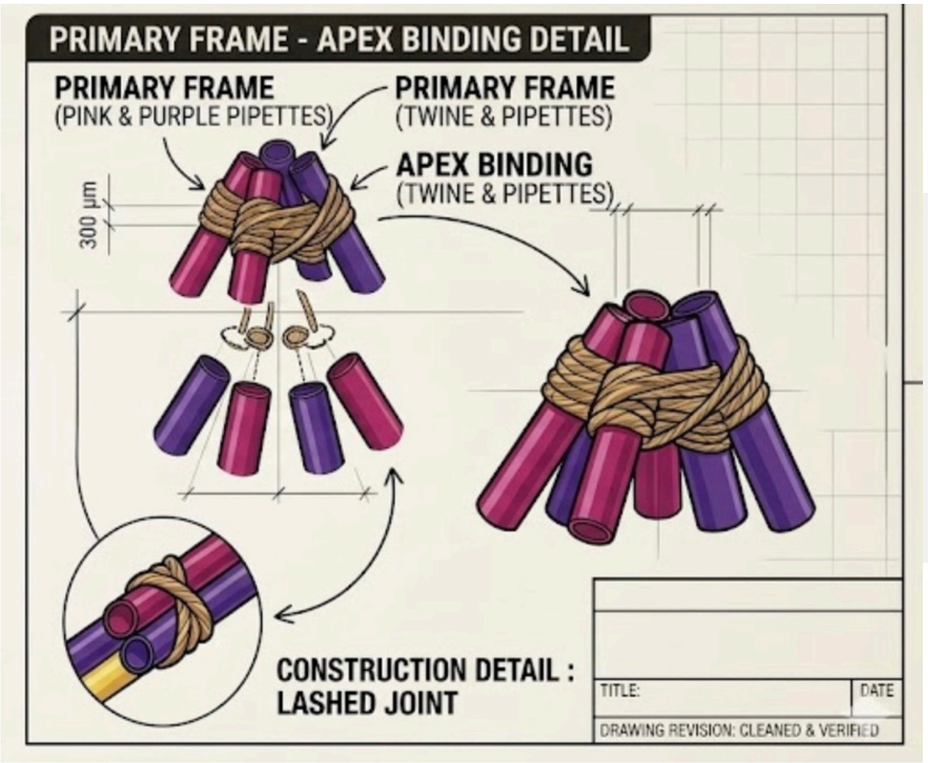

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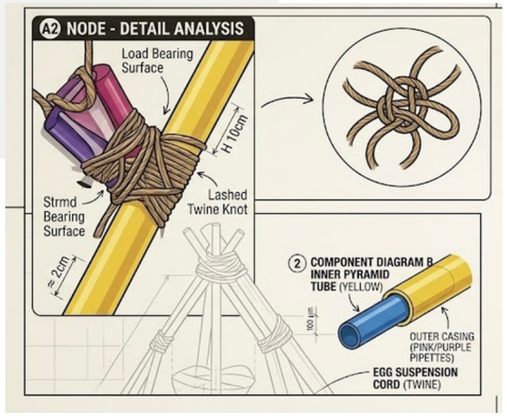
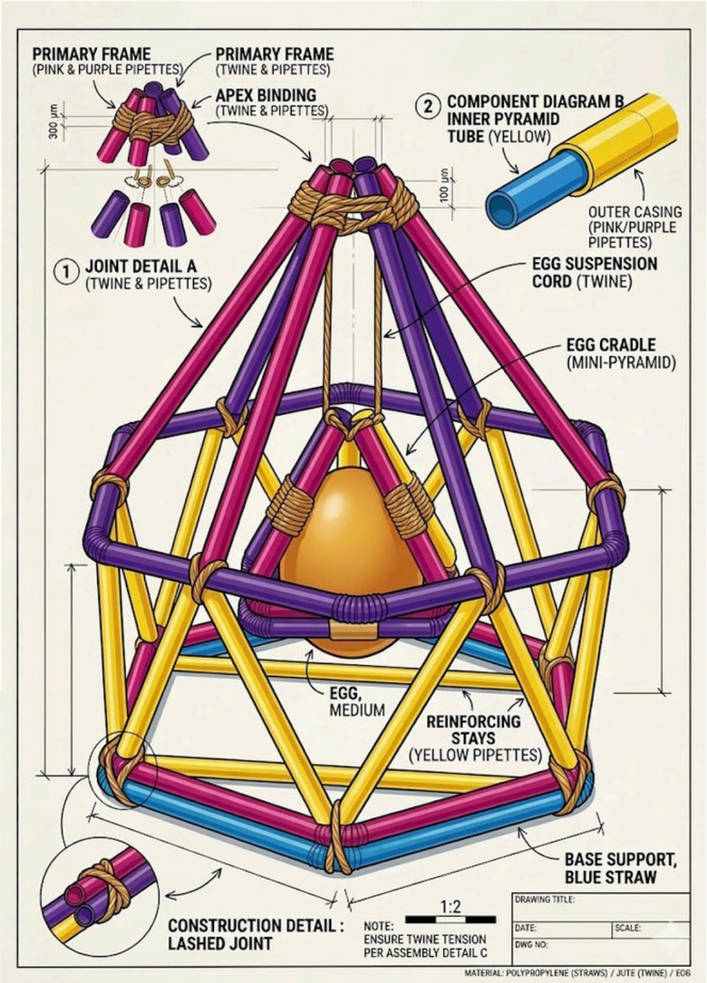


AI RESPONSE SUMMARIES

Force Flow & Physics Analysis: The AI provided a detailed explanation of how impact force travels from the ground to the bottom nodes, then through compression members (straws), and finally to the suspended egg as tension force.

The Solution: AI suggested reducing overall rigidity and creating "controlled deformation zones." By hanging the egg in a 6-point tension web, the structure allows the egg to sway slightly during the first millisecond of impact, effectively slowing down the impact over a longer duration.

05.



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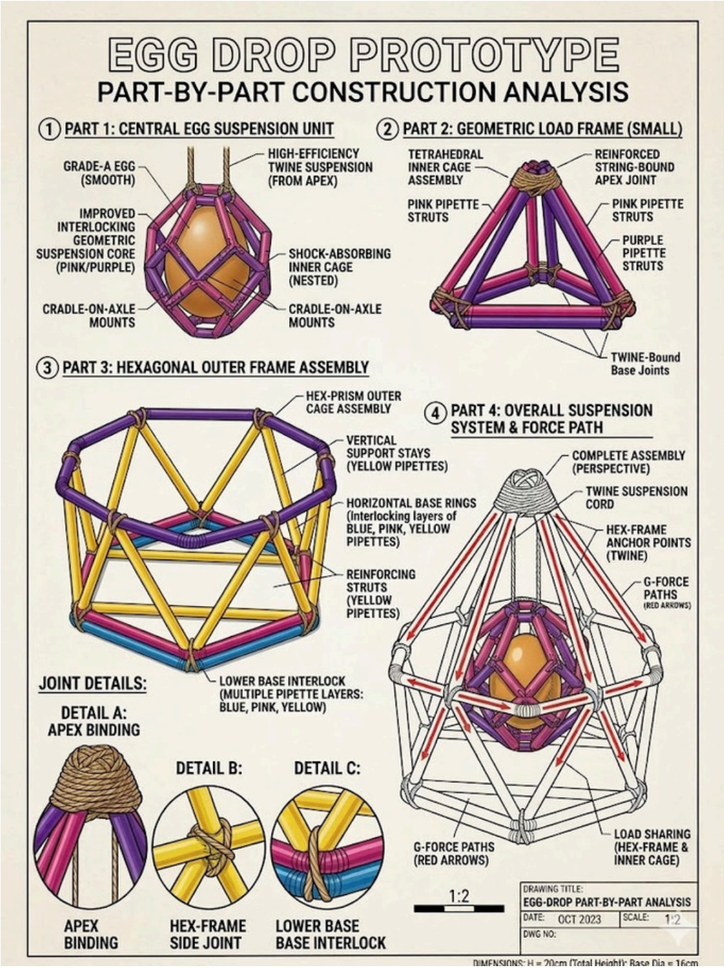
DECISION

- **Accepted:** The 6-point "Decoupled Core" suspension. Hanging the egg in a neutral zone isolated it from ground contact.

- **Rejected:** Over-complex geodesic domes. These required over 40 straws, which was impossible under the 20-straw limit.

- **Transformed:** AI suggested "crumple zones" by folding straws. Knowing this would weaken them, I transformed this into "Angled Support Members." By placing the straws at 60-degree angles, they flex outward like a spring upon impact, achieving dissipation without sacrificing integrity.



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REFLECTION

Working with AI as a "Senior Design Consultant" taught me that in industrial design, strength is found in resilience, not rigidity. I acted as the reality check for AI’s ideas, evaluating each suggestion based on the material constraints of 20 straws, 1m tape, and 2m string. HEXA-SHIELD is a testament to the fact that a successful designer does not just build shapes; they manage forces. AI provided the Why (physics), but I provided the How (craftsmanship).